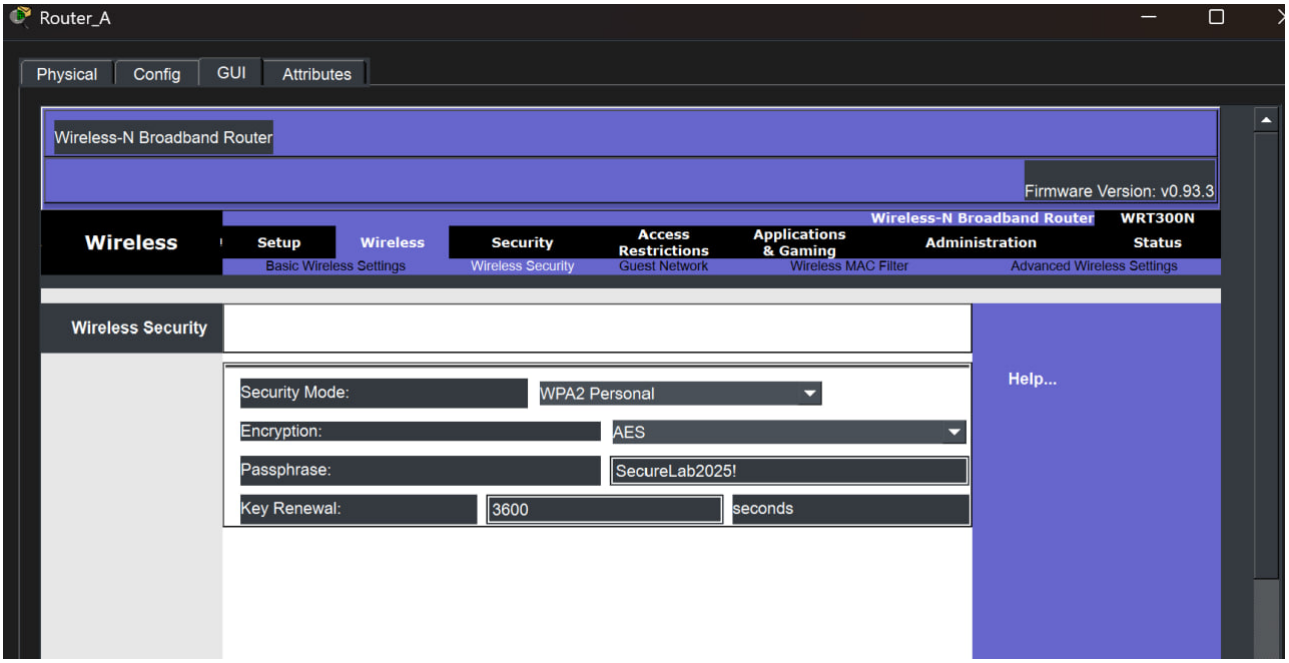
Name: Shevchenko Mariia

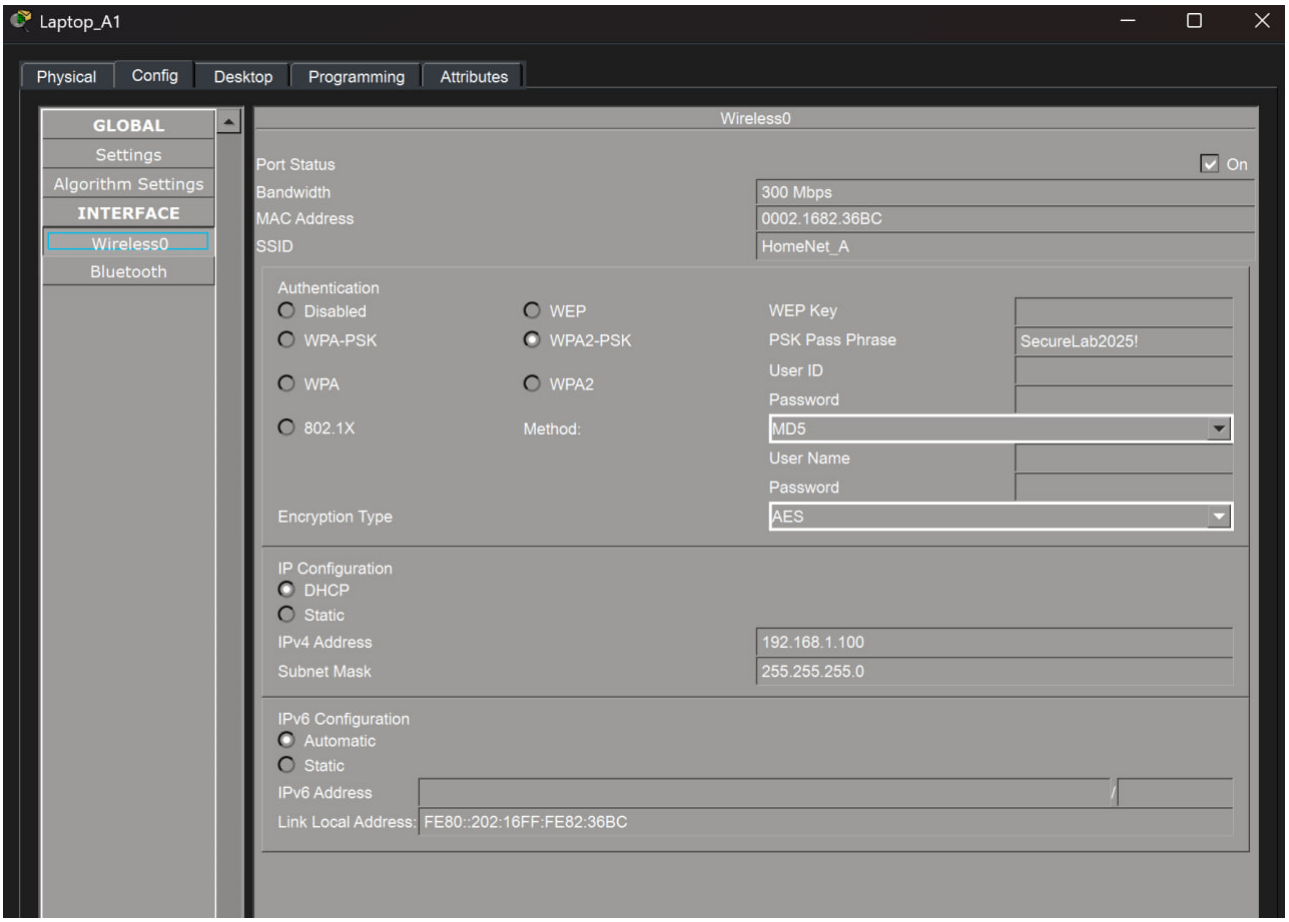
Student ID: 64594

Course: Network Security

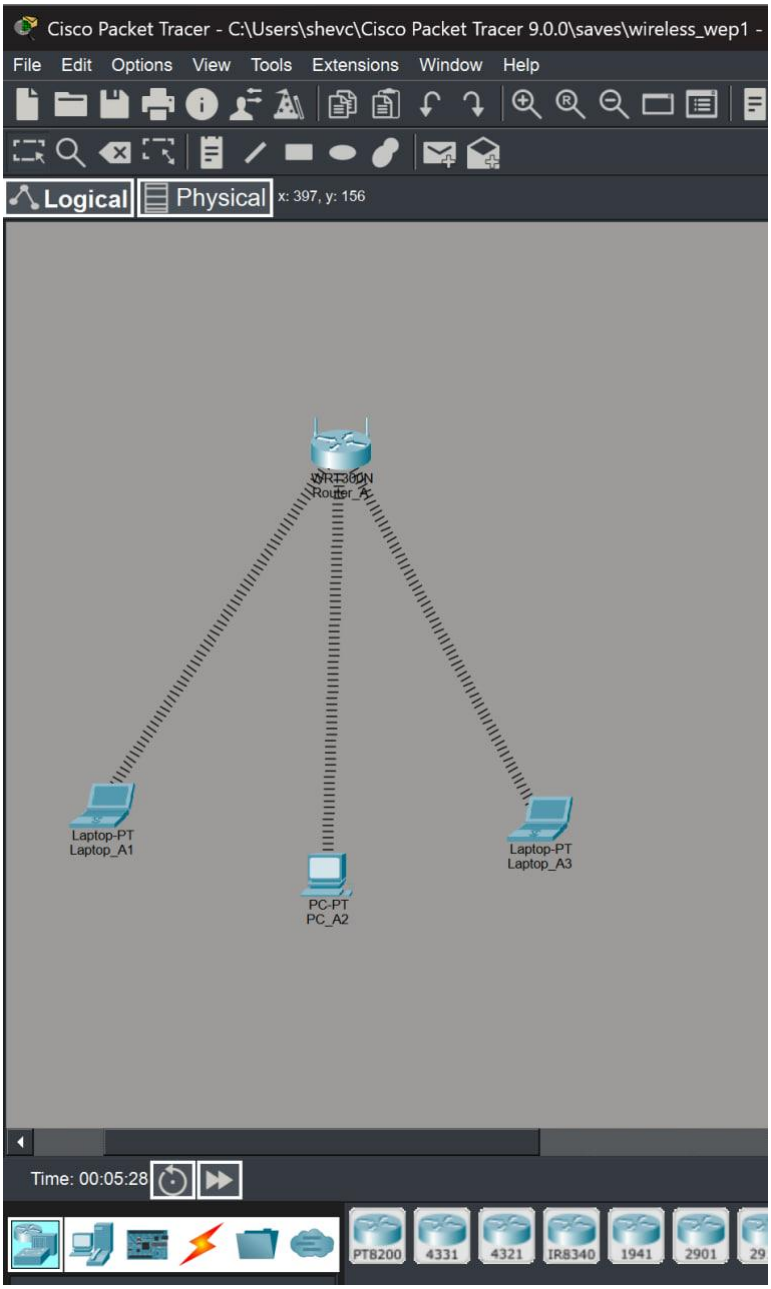
Date: 06.06.2026

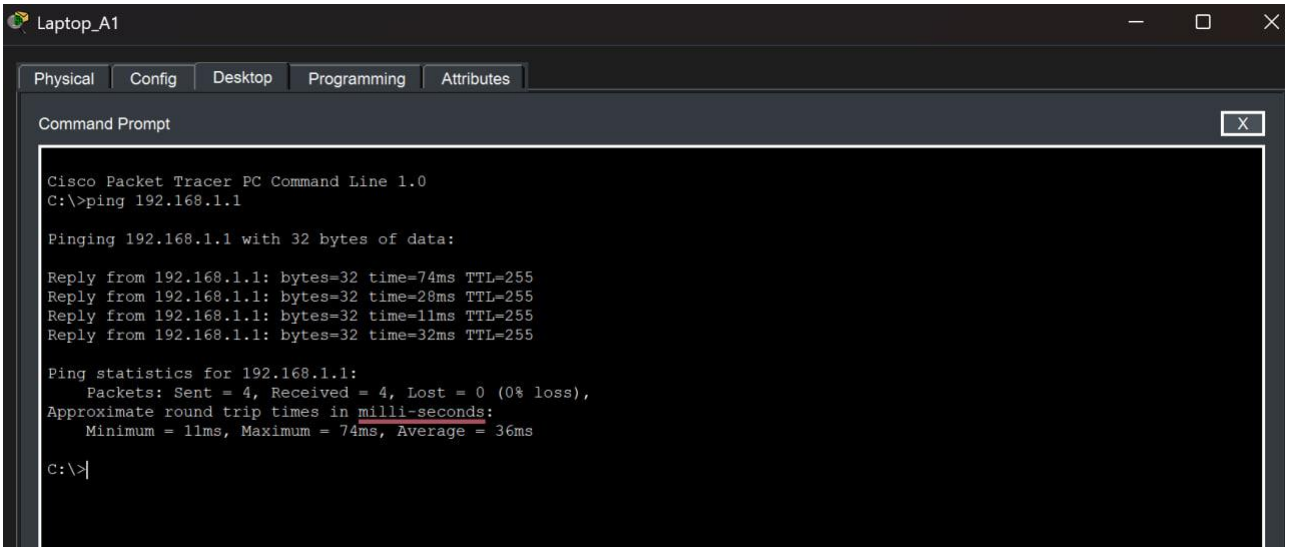


Description: This screenshot displays the security setup for Network A on Router_A. The router is configured to use WPA2-Personal security with strong AES encryption. A pre-shared passphrase has been established to secure local client access without requiring a backend server.



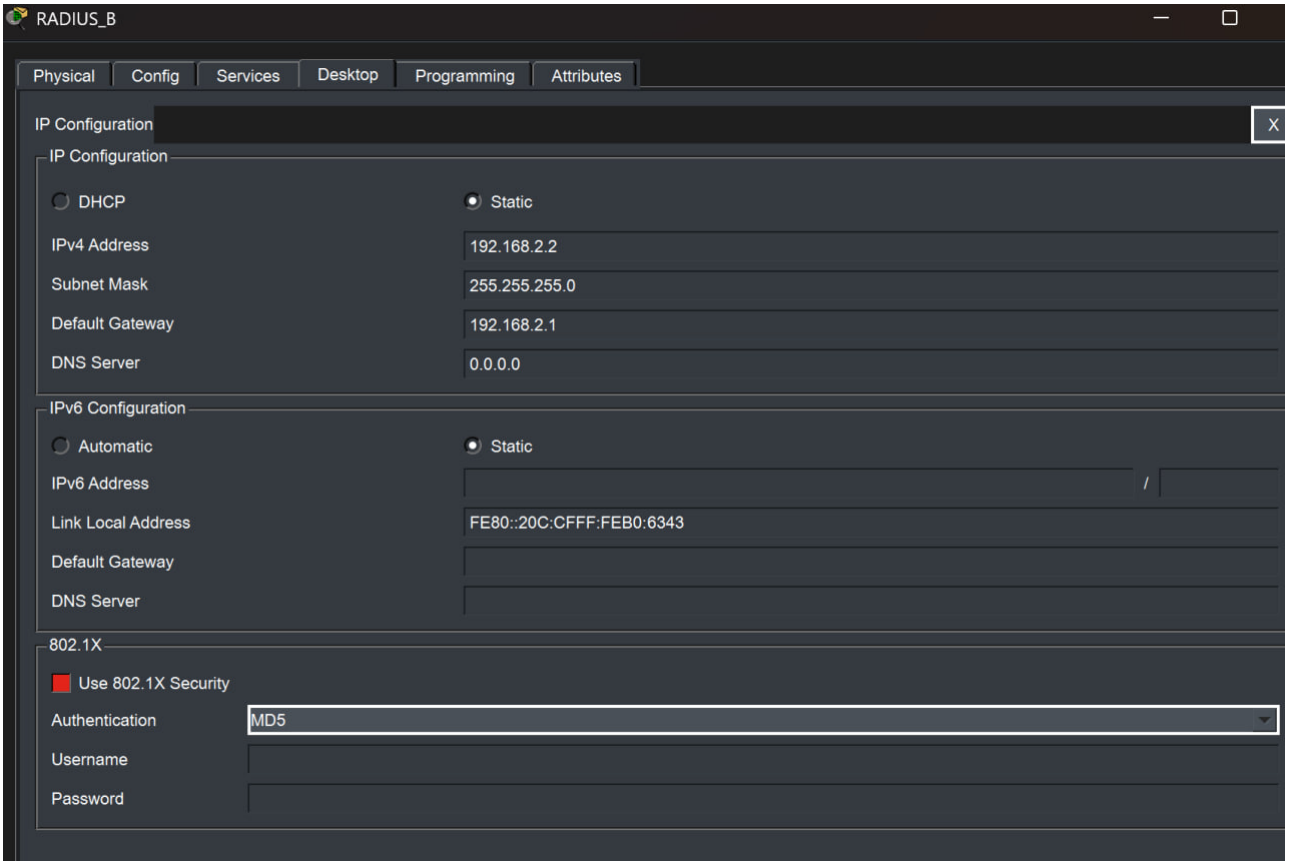
Description: This figure shows the client-side profile configuration for a device connecting to Network A. The profile uses the matching Service Set Identifier SecureNet_A. The status display confirms that the client has established a secure link to the home router.

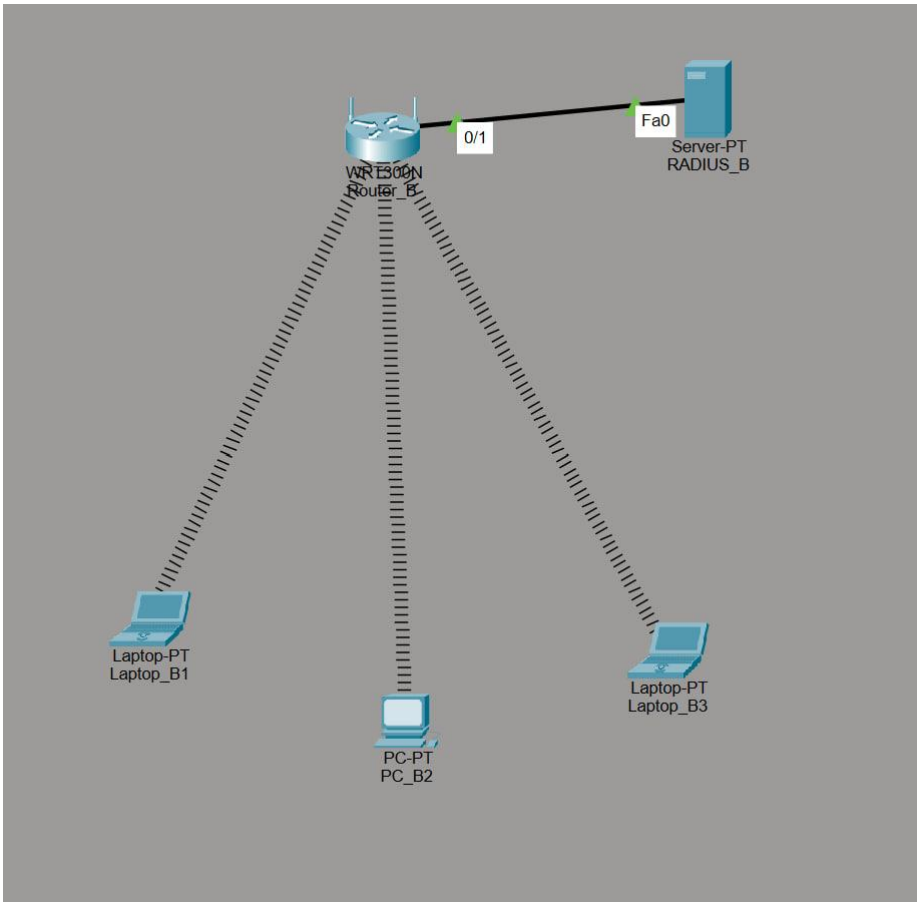




Description: This interface captures the successful network diagnostic test for Network A. The client machine has correctly obtained an IP address dynamically through DHCP. Running a

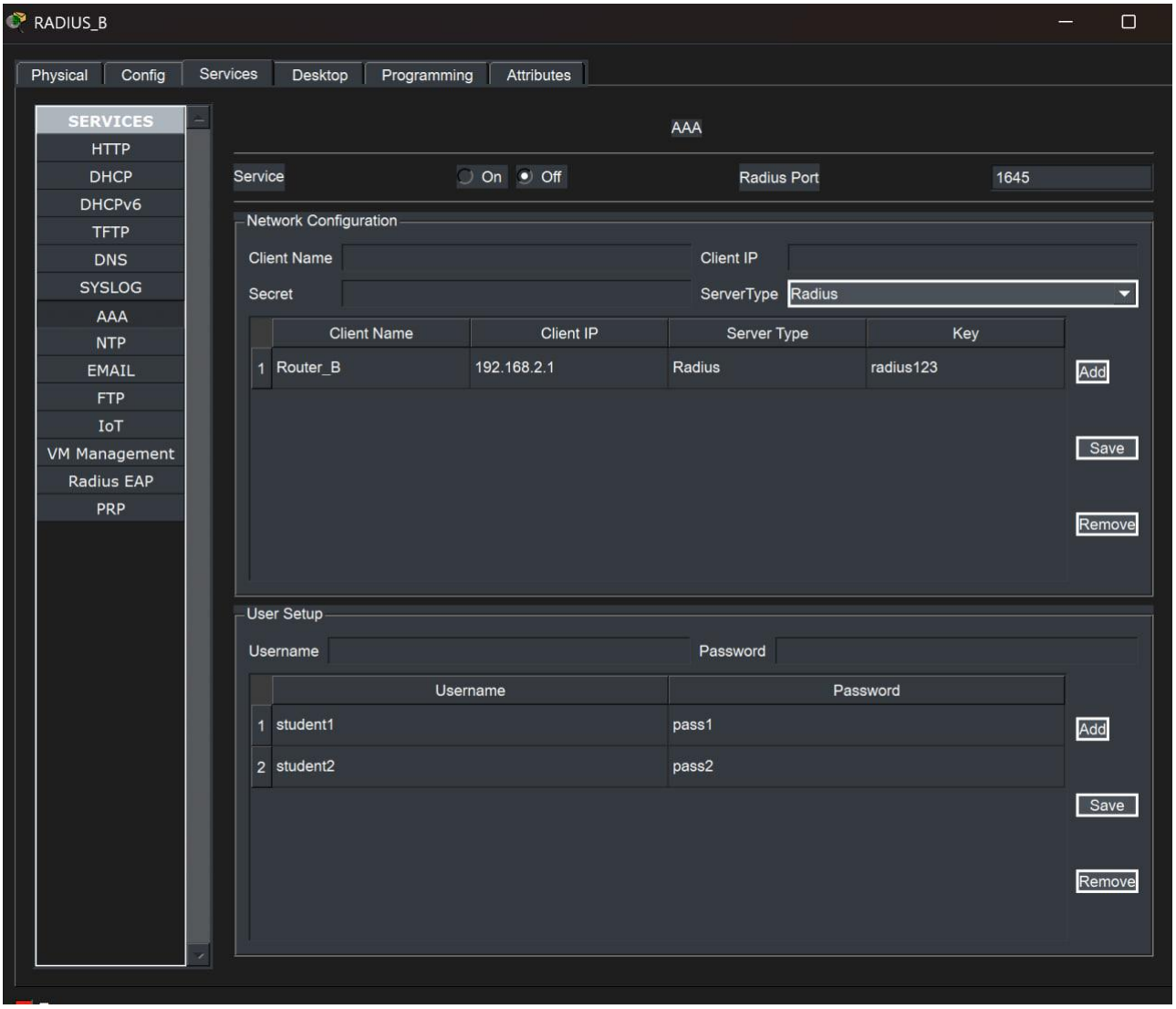
continuous ping to the gateway address 192.168.1.1 returns instant replies, verifying stable local data communication.



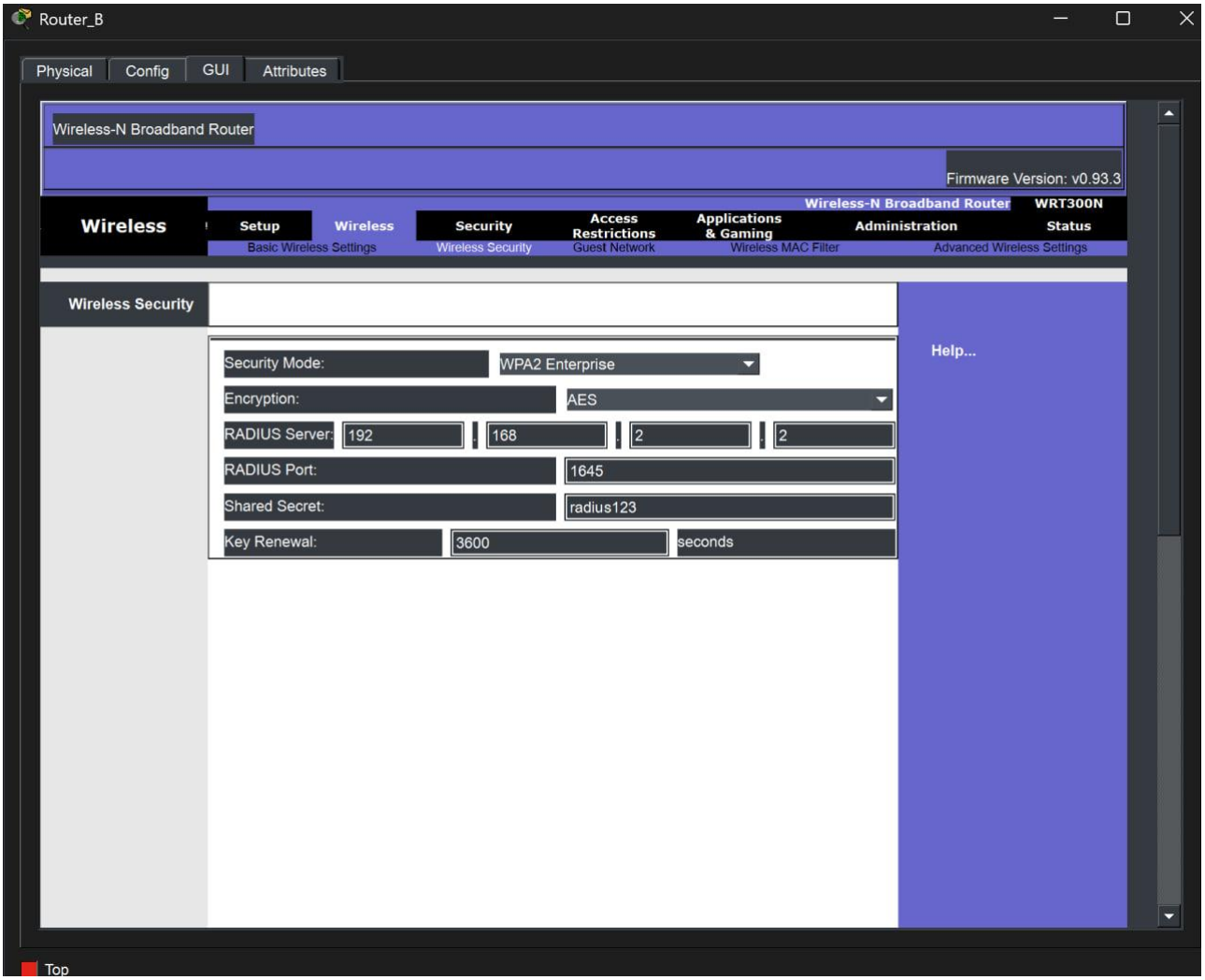


Description: This image showcases the configuration page of the standalone RADIUS authentication server. The network authentication, authorization, and accounting (AAA) service is

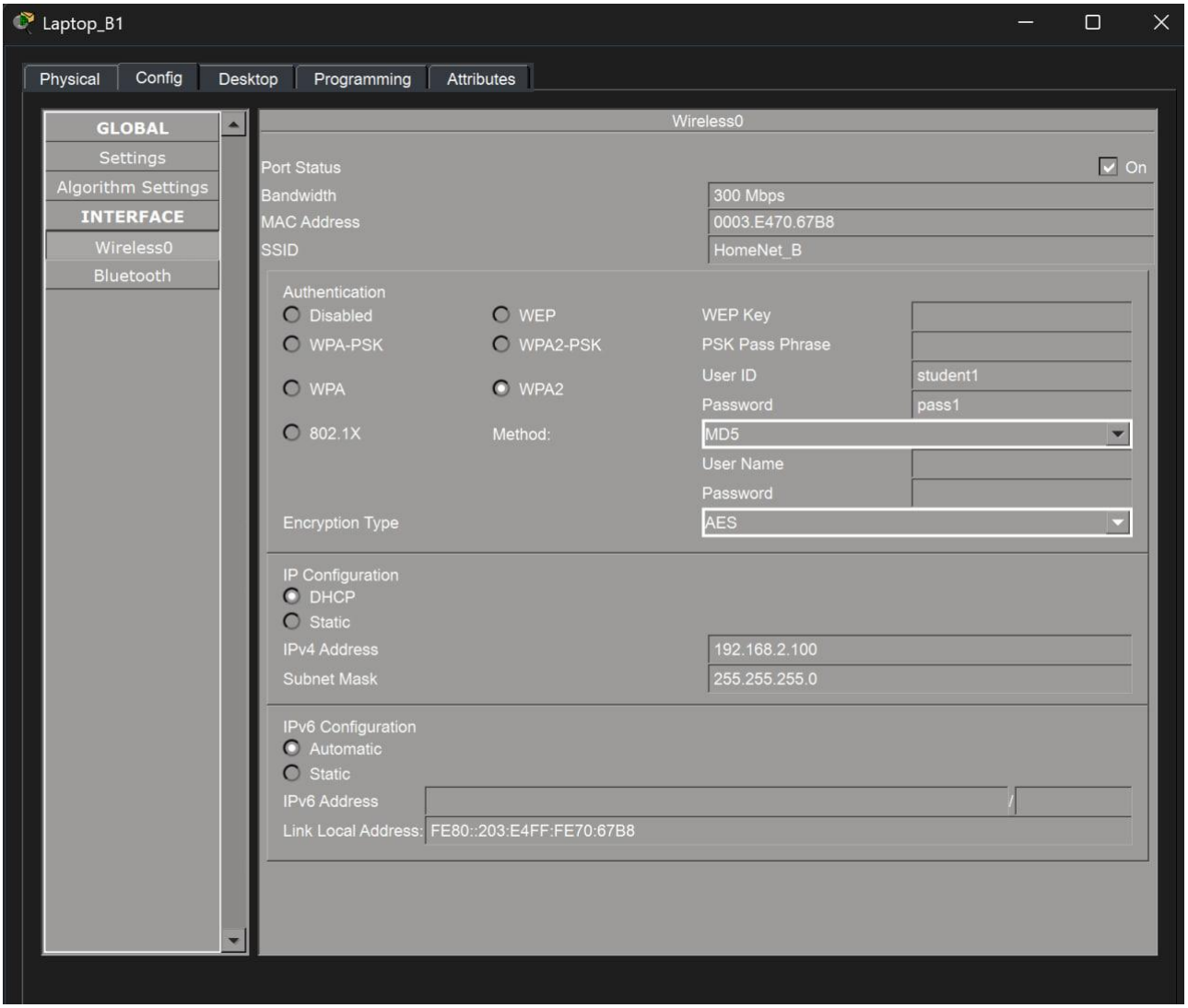
active. Router_B has been officially registered as a trusted client using its interface IP address 192.168.2.1 along with a unique shared secret key.



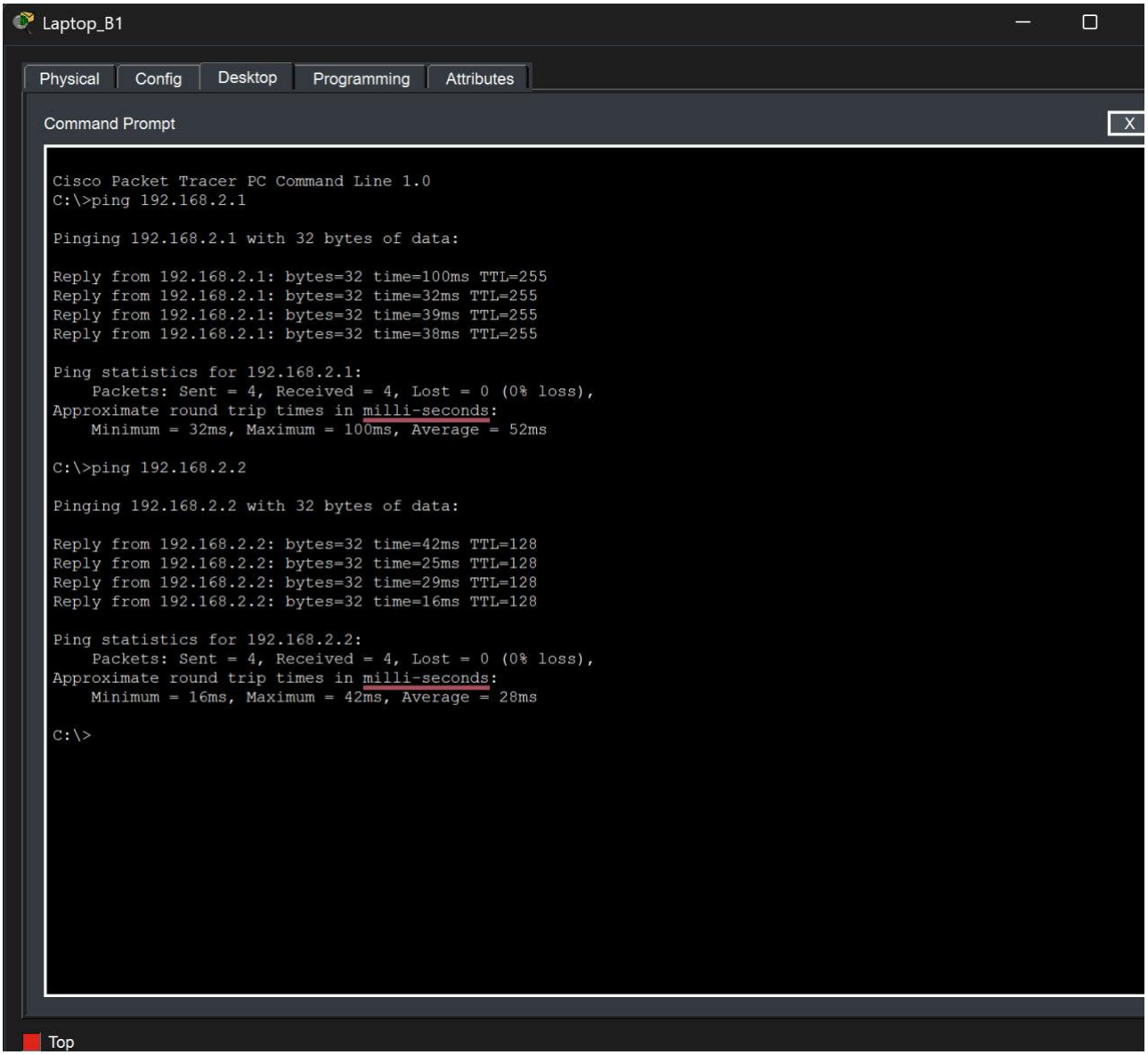
Description: This screenshot shows the AAA services page on the RADIUS server with the service turned on. It confirms that `Router_B` is successfully registered as a client and that the two individual user accounts have been added to the database.



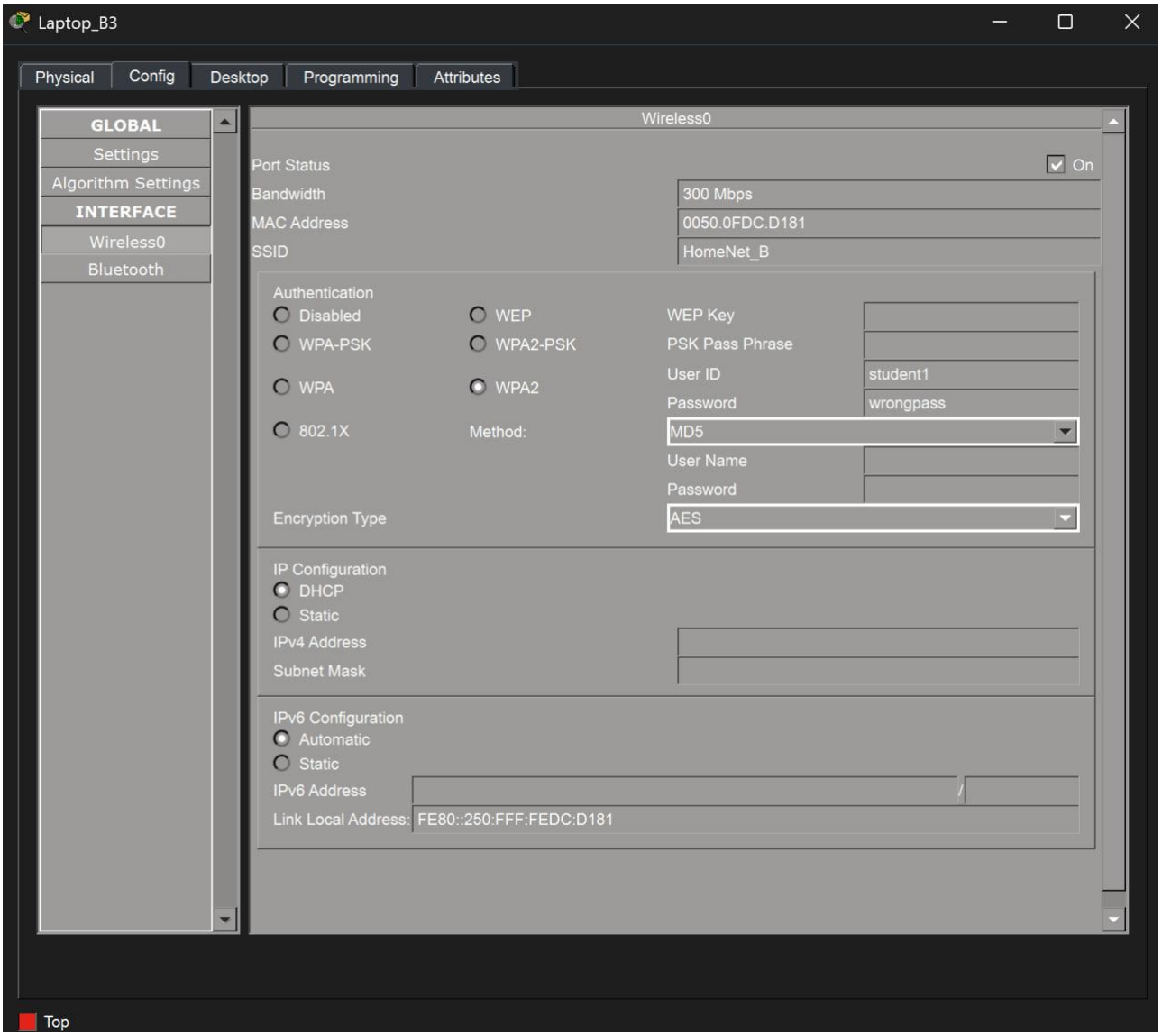
Description: This screenshot shows the wireless security settings for Router_B. The security mode is set to WPA2-Enterprise with AES encryption to protect data. It is configured to send login requests to the RADIUS server at IP address 192.168.2.2 using the shared secret password.



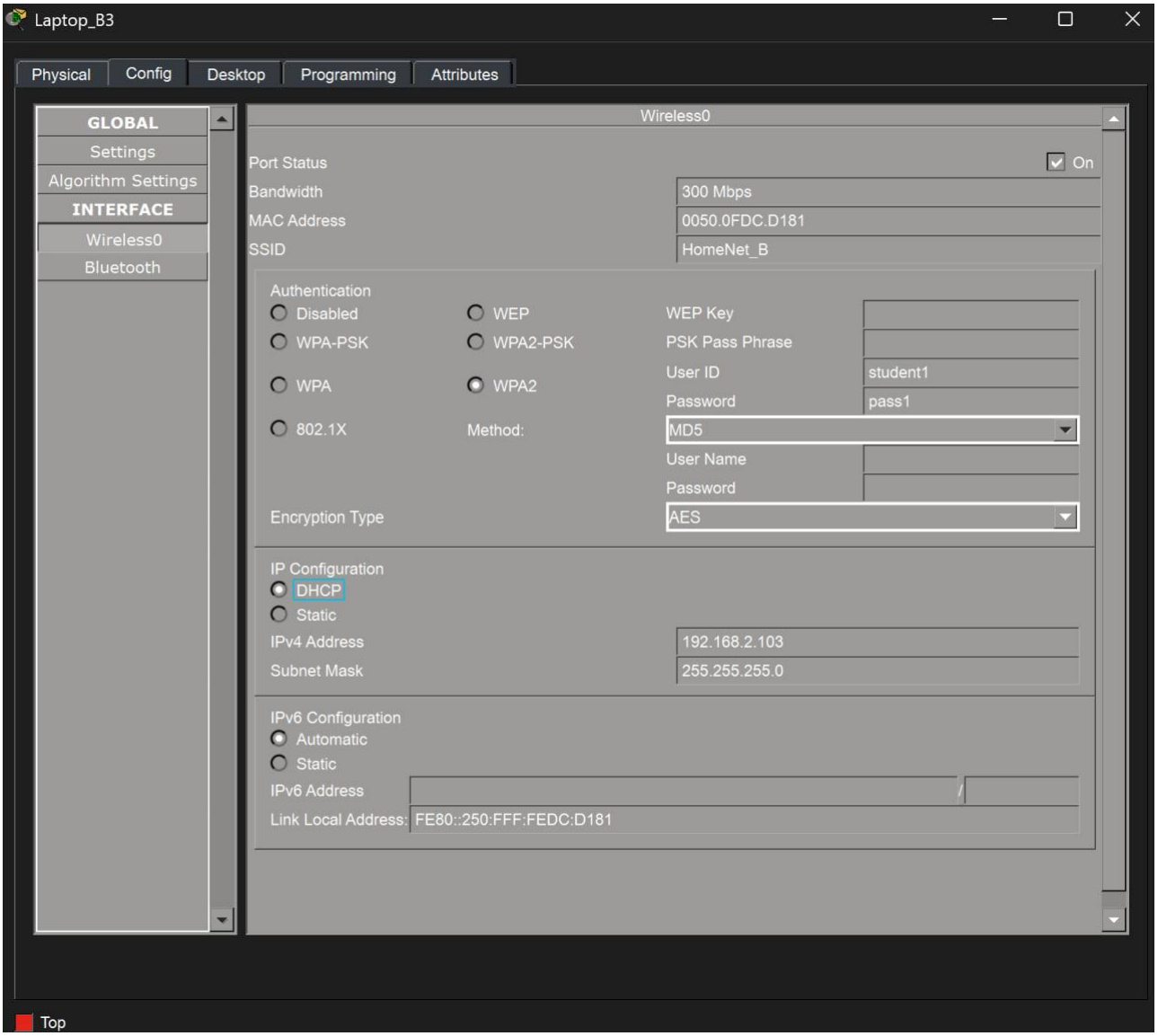
Description: This screenshot shows the wireless configuration and IP status of a client laptop on Network B. The client is set to WPA2-Enterprise with a user account, has successfully linked to the router, and has received a dynamic IP address via DHCP.



Description: This screenshot shows a successful ping test from a laptop on Network B after logging in as student1. The command prompt shows that the laptop can successfully send and receive data packets from the default gateway with zero packets lost.



Description: This screenshot shows the failed connection during the negative test on Network B. Because the laptop used an incorrect password, the RADIUS server rejected the authentication request, preventing the device from linking to the wireless network or receiving an IP address.



Corrected

Comparison:

Property	WEP(lab01)	WPA2-Personal	WPA2-Enterprise
Cipher used	RC4	AES	AES
How is the key/credential shared?	one shared key	one shared passphrase	individual user login credentials
Per-user identity?	No	No	Yes
Can you revoke ONE user?	No	No	Yes
Needs a RADIUS server?	No	No	Yes
Suitable for real use today?	No	Yes	Yes

WPA2-Personal and WPA2-Enterprise use the same AES encryption. In your own words, what problem does Enterprise solve that Personal does not?

WPA2-Enterprise eliminates the risk of a single, shared network password by requiring unique login credentials for every user. This ensures that if one user account is compromised, the rest of the network remains secure.

In the negative test, RADIUS rejected a wrong password. Explain how this delivers the "authentication" goal from Session 2 that WEP could not.

The negative test proves authentication because the RADIUS server actively verifies a user's specific identity before allowing them to connect. WEP failed at this because it only checked for a static, shared key without verifying who was actually trying to access the network.

A company has 500 employees. Give one concrete reason WPA2-Enterprise is better than a single shared passphrase when an employee leaves.

With WPA2-Enterprise, IT can instantly delete or disable just the departing employee's account on the central server in seconds. If using WPA2-Personal, IT would have to change the password on the router and manually update all 499 remaining employee devices.

What role does the RADIUS server play in the 802.1X model (supplicant / authenticator / authentication server)? Which device is which in your topology?

In the 802.1X model, the laptop acts as the Supplicant (client requesting access), Router_B acts as the Authenticator (the gatekeeper), and the RADIUS server acts as the Authentication Server (the validator).